The Cheon–Hwang Sub-Dittert Conjecture, in Full

one capacity chain for $3 \le k \le n-1$, the two endpoint lines,
and the equality case at every cell

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Draft — 3 August 2026

Abstract. Cheon and Hwang conjectured in 1992 that $E_k(r) + E_k(c) - P_k(A) \le 2 - k!/n^k$ for every nonnegative $n \times n$ matrix of total sum n and every $1 \le k \le n$, with equality only at J_n/n. The endpoint $k = n$ is Dittert's conjecture; on the doubly stochastic face the statement is the Tverberg–Friedland theorem, so all of the content sits off that face. We assemble a proof at every cell (k, n) with $2 \le k \le n$, inequality and equality case alike, and we grade every line of the assembly separately because the lines do not have the same strength.

One mechanism carries the interior. Border A into a $(2n-k) \times (2n-k)$ matrix whose permanent is $((n-k)!)^2 \sigma_k(A)$, price its permanent below by its capacity, and pay the price with an entropy witness written down in closed form from A itself. The engine of that step is refereed literature and is stated as such: Gurvits' capacity theorem, the sharp univariate extraction of Csikvári–Schweitzer and Brändén–Leake–Pak, and the Friedland–Gurvits derivation of the Friedland–Tverberg inequality. What is new here is the composition off the face: a closed form for the unrefined constant showing that its frontier is the single line $k = n-1$ (Theorem F); an exact arithmetic identity showing that repricing the $n-k$ identical border rows as one degree-$(n-k)$ extraction pays the shortfall to the last digit at every (k, n); and a strictness theorem giving $\Phi_k(A) \le 2 - \gamma - (1-\theta)D(A)$ with $1 - \theta \ge 0.7308$, so equality forces the doubly stochastic face, where Friedland's 1982 theorem finishes it. That covers $3 \le k \le n-1$ but four small cells, which the $k = 3$ and $k = 4$ lines and one anchor certificate cover.

The two endpoint lines are stated with their real provenance. At $k = 2$ and $k = 3$ the results are kernel-checked in Lean 4. At $k = n$ the line is a chain of mixed grade — refereed at $n \le 3$, our own independent certificates at $n = 4, 5$ (independent confirmation, not priority), and re-derived unrefereed preprints for $n \ge 6$ — and the assembled claim inherits that grade where it is made. We prove that the capacity chain cannot reach $k = n$, and reduce that line to a quantitative van der Waerden stability estimate which does not exist in the literature.

1 The conjecture, and what is claimed

Let $K_n = \left\{ A \in \mathbb{R}^{n \times n} : A_{ij} \ge 0, \sum_{i,j} A_{ij} = n \right\}$, and for $A \in K_n$ let r and c be its vectors of row and column sums. Write $\sigma_k(A)$ for the sum of the permanents of all $k \times k$ submatrices of A — rows chosen from one k-subset, columns from another, independently — and put

$$E_k(v) = \frac{e_k(v)}{\binom{n}{k}}, \quad P_k(A) = \frac{\sigma_k(A)}{\binom{n}{k}^2}, \quad \gamma(n,k) = \frac{k!}{n^k}, \quad \Phi_k(A) = E_k(r) + E_k(c) - P_k(A), \tag{1}$$

with e_k the elementary symmetric function. Throughout, Ω_n is the doubly stochastic polytope, $m = n - k$, and

$$D(A) = 2 - E_k(r) - E_k(c) \ge 0 \tag{2}$$

is the *line-sum deficit*; $D \ge 0$ and $D(A) = 0$ if and only if $A \in \Omega_n$, by Maclaurin's inequality applied to r and to c (this needs $k \ge 2$).

The conjecture (Cheon and Hwang [1], 1992). For every $A \in K_n$ and every $1 \le k \le n$, $\Phi_k(A) \le 2 - \gamma(n,k)$, with equality only at $A = J_n/n$.

At $k = n$ one has $\sigma_n = \text{per}$ and $E_n(v) = \prod_i v_i$, so the statement reads $\prod_i r_i + \prod_j c_j - \text{per}(A) \le 2 - n!/n^n$: *Dittert's conjecture* verbatim, Conjecture 28 of the Cheon–Wanless survey [3], open in the refereed literature at every $n \ge 4$. At $k = 1$ the inequality is an identity *on* K_n, since the difference is a constant multiple of $\sum_{ij} A_{ij} - n$; that case is excluded everywhere below. On Ω_n the deficit (2) vanishes and the conjecture becomes $\sigma_k(A) \ge \binom{n}{k}^2 \gamma(n,k)$ — the theorem conjectured by Tverberg [7] and proved by Friedland [8], whose $k = n$ case is the van der Waerden conjecture settled by Egorychev [9] and Falikman [10]. *So the entire content of Cheon–Hwang lies off the doubly stochastic face*, and that is the single fact organising this paper.

The known cases before this work were $k \le 2$ at every n; every k at $n \le 3$; $k = n$ at $n = 2$ (Sinkhorn [4]) and $n = 3$ (Hwang [5]); and the line $k = 3$ at every $n \ge 4$, together with $k = 4$ at every $n \ge 5$, from the companion paper [22]. The intermediate range $2 < k < n$ had otherwise gone untouched.

Theorem 1 (the conjecture, at every cell). For every $n \ge 2$, every $2 \le k \le n$ and every $A \in K_n$,

$$\Phi_k(A) \le 2 - \frac{k!}{n^k}, \tag{3}$$

with equality if and only if $A = J_n/n$.

The grade of this statement is composite, and the composite is only as strong as its weakest line. The range $3 \le k \le n-1$ is §2: exact-verifier grade on refereed imports. The lines $k = 2$ and $k = 3$ are kernel-checked in Lean 4; the line $k = 4$ and the four exceptional cells are exact-verifier grade (§3). **The line $k = n$ is** `[P]` **for $n \ge 6$** — unrefereed preprints, every displayed inequality re-derived here in exact arithmetic, but not refereed (§4). (3) therefore carries `[P]` wherever it is quoted as a statement about the whole plane.

1.1 How to read the grades

Five layers are used, and no claim is allowed to inherit a neighbour's layer.

layer	what it means here
kernel-checked	proved in Lean 4 at a named commit, no `sorry`, no `native_decide`, `#print axioms` returning a subset of `[propext, Classical.choice, Quot.sound]`
`[V]`	verified in exact arithmetic over $\mathbb{Q}$ end to end, by a standalone verifier carrying fault-injection controls that must fire on mutated input and stay silent on clean input
`[R]`	refereed literature, quoted and used as stated
`[P]`	unrefereed public source, re-derived here in exact arithmetic — the mathematics was checked, the refereeing was not done
`[I]`	measured at floating-point grade; scouting only, and load-bearing nowhere in this paper

Table 1: The support layers. Exact verification and machine checking are different things from refereeing, and weaker ones. Nothing in this paper is refereed, including this paper.

1.2 The map of the proof

region	what carries it	grade
$3 \le k \le n-1$, minus four cells	the capacity chain: §2, Theorem 2	`[V]` on `[R]`
$k = 2$, every $n \ge 2$	a manifest sum of squares; §3.1	kernel-checked
$k = 3$, every $n \ge 4$	one certificate family in $\mathbb{Q}(n)$; §3.2, [22]	kernel-checked
$k = 3$, $n = 3$	Hwang 1987 [5] (this is Dittert at $n = 3$)	`[R]`
$k = 4$, every $n \ge 4$	(4,4) certificate, five anchors, one collar theorem; §3.3	`[V]`
$(k, n) = (5, 6)$	anchor certificate; §3.4	`[V]`
$k = n$, $n \le 3$	Sinkhorn [4], Hwang [5]	`[R]`
$k = n$, $n = 4, 5$	our own anchor certificates; §5.3	`[V]`
$k = n$, $n \ge 6$	three unrefereed preprints, re-derived; §4	`[P]`

Table 2: Coverage of the (k, n) plane, $2 \le k \le n$. The four cells excluded from the capacity range are $(k, n) = (3, 4), (3, 5), (4, 5), (5, 6)$, and each is covered by a row below it.

2 The capacity chain

This section proves the conjecture, in full, on

$$R = \{(k, n) : 3 \le k \le n-1\} \setminus \{(3, 4), (3, 5), (4, 5), (5, 6)\}. \tag{4}$$

One mechanism does all of it. The engine is imported and is refereed; the composition off the face, the frontier analysis, the exact payment and the strictness theorem are the contribution.

2.1 The bordered matrix and its capacity

For $2 \le k \le n$ put $m = n - k$, $N = 2n - k$, and

$$M_k(A)=\begin{pmatrix} A & J_{n\times m}\\ J_{m\times n} & 0_{m\times m}\end{pmatrix},\quad \mathrm{per}(M_k(A))=(m!)^2\sigma_k(A). \tag{5}$$

In a permanent term of $M_k(A)$ the m all-ones columns can only be covered by A-rows and the m all-ones rows only by A-columns, since the corner block vanishes; so exactly k A-rows meet A-columns, those index sets are the pair (α,β), and the two leftover matchings contribute $m!$ each. Write $p_M(x)=\prod_i\left(\sum_j M_{ij}x_j\right)$, a product of linear forms with non-negative coefficients, hence H-stable and homogeneous of degree N in N variables, with $\mathrm{per}(M)=\partial^N p_M/\partial x_1\cdots\partial x_N(0)$, and

$$\mathrm{cap}(p)=\inf_{x>0}\frac{p(x)}{x_1\cdots x_N}. \tag{6}$$

> **Lemma C (the capacity is constant on the face).** For every $A\in\Omega_n$ and every $1\le k\le n$,
>
> $$\mathrm{cap}(M_k(A))=\mathrm{cap}_0:=\frac{n^{2n-k}}{k^k}, \tag{7}$$
>
> attained at $x^*=(1,\dots,1,1/k,\dots,1/k)$ with n ones and m copies of $1/k$.
>
> [V]. The Legendre-type objective $\log p_M(e^y)-\sum_j y_j$ is convex, so the displayed stationary point is a global minimiser; both stationarity computations use only $r\equiv c\equiv 1$.

Two consequences are worth stating at once. First, $d\log\mathrm{cap}/dA_{ij}=k/n$ at every (i,j) on the face, so the directional derivative of $\mathrm{cap}(M_k(A))$ along any direction inside K_n vanishes on Ω_n: *the capacity is flat on the face.* Second, and for the same reason, any bound of the form $\sigma_k\ge c\cdot\mathrm{cap}$ has *no slack at all* at J_n/n and must therefore be exactly tight there or fail outright. That is the whole difficulty of the constant, and it is why the frontier of §2.3 is a line rather than a region.

2.2 The engine, and where it comes from

> **Provenance, stated up front.** The three inputs below are refereed or published literature and are not our results. Each was independently re-derived here in exact arithmetic — that is a check on our use of them, not a claim on them.
>
> **(E1) Gurvits' capacity theorem** [11]. For p homogeneous, H-stable, of degree N in N variables with non-negative coefficients, with $C_i=\min(\deg_p(i),i)$,
>
> $$\frac{\partial^N p}{\partial x_1\cdots\partial x_N}(0)\ge\mathrm{cap}(p)\cdot\prod_{2\le i\le N}((C_i-1)/C_i)^{C_i-1}. \tag{8}$$
>
> **(E2) The sharp univariate extraction.** For $a_1,\dots,a_n\ge 0$ and $f(S)=\prod_i(a_i+S)=\sum_t c_tS^t$, and $0\le m\le n$,
>
> $$c_m\ge G(n,m)\cdot\inf_{S>0}f(S)/S^m,\quad G(n,m)=\frac{\binom{n}{m}m^m(n-m)^{n-m}}{n^n}\sim\sqrt{\frac{n}{2\pi m(n-m)}}, \tag{9}$$
>
> with equality at $a=(1,\dots,1)$. This is Csikvári–Schweitzer [13], Lemma 2.8 (attributed there to Gurvits [15]), and Brändén–Leake–Pak [14], Corollary 5.9, where sharpness is proved. It is *not* ours.
>
> **(E3) The descent mechanism.** Multiplying by $L(x)^{n-k}$, which does not decrease capacity, to descend from the $m=0$ case *is* the Friedland–Gurvits proof of the Friedland–Tverberg inequality [12], Theorem 3.1 and Corollary 3.2 (2006). §2.4 below is, in its mechanism, a rediscovery of that 2006 theorem.

Applying (E1) to p_M with the labelling that parks the m border columns at the top labels — the optimal labelling, since $((d-1)/d)^{d-1}$ decreases — gives the *unrefined* constant

$$\mathrm{per}(M_k(A))\ge C_{\mathrm{ref}}(n,k)\,\mathrm{cap}(M_k(A)),\quad C_{\mathrm{ref}}(n,k)=\frac{n!}{n^n}((n-1)/n)^{(n-1)(n-k)}, \tag{10}$$

valid for every $A\ge 0$. Write $\rho_{\mathrm{ref}}(n,k)=C_{\mathrm{ref}}\,\mathrm{cap}_0/\left((m!)^2\binom{n}{k}^2\gamma\right)$ for the ratio of the certified value at J_n/n to the true one. By the flatness just noted, the chain runs at (k,n) only if $\rho_{\mathrm{ref}}\ge 1$, and $\rho_{\mathrm{ref}}\le 1$ always.

2.3 The frontier of the unrefined constant is a line

> **Theorem F (the frontier, exactly).** For every $n\ge 3$ and every $1\le k\le n-1$,
>
> $$\frac{\rho_{\mathrm{ref}}(n,k)}{\rho_{\mathrm{ref}}(n,k+1)}=\frac{(1+1/k)^k}{(1+1/(n-1))^{n-1}}. \tag{11}$$
>
> Since $x\mapsto(1+1/x)^x$ is strictly increasing, ρ_{ref} is strictly increasing in k and flat from $n-1$ to n; with $\rho_{\mathrm{ref}}(n,n)=1$,
>
> $$\rho_{\mathrm{ref}}(n,n-1)=1,\quad \rho_{\mathrm{ref}}(n,k)=\prod_{i=k}^{n-2}\frac{(1+1/i)^i}{(1+1/(n-1))^{n-1}}<1\quad\text{for every } k\le n-2. \tag{12}$$

> `[V] graded_verify_capfrontier.py`: all 29 checks pass, all 10 fault-injection controls fire. The ratio identity is verified over $\mathbb{Q}$ on 3159 cells at $n = 4...80$ and the product form at $n = 3...49$; the closed form reproduces the earlier measured table digit for digit on all 21 of its rows. *This is ours.*

Theorem F converts a measured table and an asymptotic guess into a theorem, and the asymptotic $1-\rho_{\mathrm{ref}} \sim (n-k)^2/(4n^2)$ falls out of it. It also prices what a wider region would cost: reaching $k = n-j$ requires the Gurvits constant to improve at the bordered degree profile by exactly $1/\rho_{\mathrm{ref}}(n, n-j)$ — a factor 1.00600 at $(n, j) = (10, 2)$, tending to 1 like $j^2/(4n^2)$ at fixed j, but tending to $1/\sqrt{ce^{1-c}} > 1$ along $k = cn$. So no limiting argument delivers the interior, and a constant fraction of the plane costs a genuine constant-factor strengthening of a van der Waerden-type bound. *One structural slack is visible*: (E1) does not use that the m border rows of $M_k(A)$ are identical linear forms.

2.4 The exact payment

That slack is the whole of the gap, and paying it closes the gap exactly — not asymptotically. The border variables enter p_M only through their sum:

$$p_M(x, x_{n+1}, ..., x_{n+m}) = L(x)^m \prod_{i\le n}((Ax)_i + S), \quad S = \sum_b x_{n+b}, \quad L(x) = \sum_{j\le n} x_j, \tag{13}$$

so the m-fold border derivative is $m![S^m]$: *one* coefficient extraction from *one* variable of degree n, which is exactly the situation of (9). The unrefined constant instead charges $((n-1)/n)^{n-1}$ once per border variable — it spends m single-variable steps on a single extraction, and the resulting loss e^{-m} against $\sim 1/\sqrt{m}$ is entirely an artefact of that choice.

> **Theorem G′ (refined constant).** For every $A \ge 0$,
>
> $$\mathrm{per}(M_k(A)) \ge C_{\mathrm{new}}(n,k)\,\mathrm{cap}(M_k(A)), \quad C_{\mathrm{new}}(n,k) = \frac{n!}{n^n}\cdot\frac{m!\,G(n,m)}{m^m}, \quad m = n-k. \tag{14}$$
>
> **Theorem H′ (the collapse).** $C_{\mathrm{new}}(n,k)\,\mathrm{cap}_0 = (m!)^2\binom{n}{k}^2\gamma(n,k)$ *exactly*, i.e. $\rho_{\mathrm{new}}(n,k) = 1$ at *every* k; equivalently $C_{\mathrm{new}}/C_{\mathrm{ref}} = 1/\rho_{\mathrm{ref}}$ identically.
>
> **Mechanism imported** — (E2) and (E3) of §2.2; the honest description of Theorem G′ is that it is Friedland–Gurvits [12] applied to this family. What is ours is the composition and the bookkeeping identity below. `[V] graded_verify_borderrows.py`: all 20 checks pass, all 7 controls fire; $\rho_{\mathrm{new}} = 1$ on every cell $2 \le k \le n$, $n = 3...44$, and $C_{\mathrm{new}} = C_{\mathrm{ref}}$ exactly at $m = 0, 1$.

The proof of Theorem H′ is one line of arithmetic, and it is worth writing out because a constant that closes to the last digit at every cell is rare. Both sides reduce, using $(n-m)^{n-m} = k^k$, to

$$n!\,m!\binom{n}{m} = (m!)^2\binom{n}{k}^2 k! \quad (= (n!)^2/k! \text{ on both sides}), \tag{15}$$

which is $\binom{n}{m} = \binom{n}{k}$ together with $\binom{n}{k} = n!/(k!m!)$. The identity is forced by the identical-rows structure: the same $m!$ that appears in (5) as the two leftover matchings appears in (9) as the extraction factor.

Two remarks, both recorded deliberately. First, $G(n,1) = ((n-1)/n)^{n-1}$ and $G(n,0) = G(n,n) = 1$, so the refinement *reduces* to (10) exactly at $m = 0, 1$ — the two lines where Theorem F says the unrefined labelling was already sharp. Nothing is claimed where nothing was owed. Second, on Ω_n Theorem H′ reads $\sigma_k(A) \ge \binom{n}{k}^2\gamma$, which is Friedland–Tverberg. That agreement is *not* corroboration: the derivation is Friedland and Gurvits' own derivation of that inequality, so the check cannot fail. It fixes the depth of the claim — the refined bound is exactly as strong as a known sharp theorem on the face — and no evidential weight is placed on it.

2.5 The witness, and the region

The chain has four steps, and all four are tight at J_n/n:

- (S1) $E_k(r) + E_k(c) = 2 - D(A)$ — the definition (2);
- (S2) $P_k(A) \ge \gamma \cdot \mathrm{cap}(M_k(A))/\mathrm{cap}_0$ — Theorems G′/H′;
- (S3) $\mathrm{cap}(M_k(A)) \ge \mathrm{cap}_0 \exp(-(m/n)\sum\chi)$ — the entropy witness;
- (S4) $(m/n)\sum\chi \le D(A)/\gamma$ — the χ-comparison.

Step (S3) is where a lower bound on an infimum is needed, and a lower bound on an infimum is a *dual feasible point*: a guess that only has to be checked, never found. Let $W \ge 0$ be doubly stochastic with $\mathrm{supp}(W) \subseteq \mathrm{supp}(M)$; then weighted AM–GM on each row, multiplied over i and using the column sums of W, gives $\mathrm{cap}(M) \ge \prod_{i,j}\left(M_{ij}/W_{ij}\right)^{W_{ij}}$ — the easy half of the classical scaling duality, proved here from scratch so that nothing external is load-bearing at this step. The point that works is written down from A itself: for $A \in K_n$ with $\max_i r_i \le n/k$ and $\max_j c_j \le n/k$, put

$$W_{ij} = (k/n)A_{ij}, \quad W_{i,n+b} = \frac{1-(k/n)r_i}{m}, \quad W_{n+a,j} = \frac{1-(k/n)c_j}{m}, \tag{16}$$

with the $m \times m$ corner zero. Its row and column sums are 1 in one line each, and evaluating the witness with $\chi(t) = (1-t)\log(1-t) + t = \sum_{p\geq 2} t^p/(p(p-1)) \geq 0$ gives exactly (S3) with χ evaluated at $(k/m)R_i$ and $(k/m)C_j$, where $R = r - \mathbf{1}$, $C = c - \mathbf{1}$.

> **The witness is exactly tight on the face.** At $R = C = 0$ every χ vanishes and the bound returns cap_0, which Lemma C says is the true value. So (S3) is not a lossy surrogate for the capacity; it *is* the capacity on Ω_n, and the loss off the face is the explicit χ-sum.
> [V] checked as an identity in the free $\mathbb{Q}$-module on $\{\log p : p \text{ prime}\}$, $n = 4...9$, every $2 \leq k < n$, at J_n/n, at permutation matrices and at off-face product points (`graded_verify_thmb.py`, all 32 checks pass, all 9 controls fire).

The hypothesis $\max(r_i, c_j) \leq n/k$ has to be earned, and it is: on the region $\{D < \gamma\}$ — the only region where anything is owed, since $D \geq \gamma$ gives (3) outright from (S1) and $P_k \geq 0$ — an exact extremal computation for $\hat{E}_k(u) = \max\{E_k(r) : r \geq 0, \sum r = n, r_1 = u\}$ pins every line sum. Step (S4) then needs three rational conditions,

$$\text{(C1) } \gamma \leq 1/12, \quad \text{(C2) } 3\gamma k^2(n-1)^2 \leq (m(k-1))^2, \quad \text{(C3) } \gamma k^2(n-1) \leq m(k-1)(1-\kappa), \tag{17}$$

with $\kappa = 3\gamma(k-2)(n-1)/(k-1)^2$. These hold on exactly the region (4).

> **The exclusion set, exactly.** (C1)–(C3) hold at every $3 \leq k \leq n-1$ except at $(k,n) = (3,4), (3,5), (4,5), (5,6)$. The cell set is infinite, so the infinite half is a lemma and not a table. Write $m = n - k$ and
>
> $$\Lambda(n,k) = \frac{\gamma k^2(n-1)}{m(k-1)}, \quad \Xi(n,k) = \frac{3\gamma k^2(n-1)^2}{(m(k-1))^2}, \tag{18}$$
>
> so that (C2) reads $\Xi \leq 1$, (C3) reads $\Lambda \leq 1 - \kappa$, and $\theta = \Lambda/(1-\kappa)$ in (19) below.
>
> **Lemma T (the tail).** For every $n \geq 10$ and every $3 \leq k \leq n-1$, the conditions (C1), (C2) and (C3) all hold, and moreover $\theta(n,k) \leq 144/955 = 0.15078... < 1$.
>
> *Proof.* Five elementary steps over $\mathbb{Q}$, (T0)–(T4), and an assembly (T5). (T0) γ is non-increasing in k, since $\gamma(n,k+1)/\gamma(n,k) = (k+1)/n \leq 1$ on $k \leq n-1$; hence $\gamma \leq \gamma(n,3) = 6/n^3$ across the strip, and $\gamma \leq h!/n^h$ with $h = \lfloor n/2 \rfloor + 1$ once $k \geq h$. (T1) $(k-1)^2 - 4(k-2) = (k-3)^2 \geq 0$, so $(k-2)/(k-1)^2 \leq 1/4$ and $\kappa \leq (3/4)\gamma(n-1) < 9/(2n^2)$; at $n \geq 10$ this gives $1 - \kappa > 191/200$, so θ is well defined and positive. (T2) Small k, $3 \leq k \leq n/2$: then $m \geq n/2$, $k-1 \geq k/2$ and $k-1 \geq 2k/3$, whence $\Lambda \leq 24k/n^3 \leq 12/n^2$ and $\Xi \leq 27\gamma \leq 162/n^3$. (T3) Large k, $n/2 < k \leq n-1$: this is the dangerous side, where m may be 1 and the $(n-k)$ in the denominator gives nothing, so the super-exponential decay of γ replaces it. With $\gamma \leq h!/n^h$ and $k - 1 \geq (n-1)/2$, $\Lambda \leq B(n) := 2n^2 h!/n^h$ and $\Xi \leq 6B(n)$. (T4) $B(n+2) \leq B(n)$ for $n \geq 8$, since $B(n+2)/B(n) \leq (n+4)(n+2)/(2n^2) \leq 1$ once $n^2 - 6n - 8 \geq 0$; with $B(10) = 18/125$ and $B(11) = 1440/14641 < 18/125$, induction along each parity class gives $B(n) \leq 18/125$ for every $n \geq 10$. (T5) Assembly at $n \geq 10$. (C1): $\gamma \leq 6/n^3 \leq 6/1000 < 1/12$. (C2): $\Xi \leq 162/1000$ by (T2) and $\Xi \leq 6B(n) \leq 108/125$ by (T3). (C3): $\Lambda \leq \max(12/100, 18/125) = 18/125$, so $\theta \leq (200/191)(18/125) = 144/955$. ■
>
> The crossover is $n_1 = 10$, and the condition that fixes it is (C2), not (C3) — the opposite of the small-n picture. (C3) alone would close from $n \geq 8$, where $B(8) = 15/32$ gives $\theta \leq 60/119 < 1$; but (T3)'s certificate for (C2) is $6B(n)$, and $6B(8) = 45/16$ and $6B(9) = 160/81$ both exceed 1. So the exact finite part is the 21 cells with $4 \leq n \leq 9$, decided cell by cell over $\mathbb{Q}$, of which exactly the four listed fail. Lemma T is a ceiling and not a value: the true maximum of θ on the tail is 0.03618... at $(k,n) = (3,10)$, a factor 4.2 under 144/955, and it falls away super-exponentially after that. What the lemma buys is reach, not sharpness; the tight corner of R is not in the tail at all, but at $(k,n) = (6,7)$ inside the finite part.
> [V] `graded_verify_strict.py` block [6]: (T0)–(T5) are each re-checked as the inequality each one is, over the 12403 cells with $n \leq 160$, together with the crossover $n_1 = 10$, the 21 exact cells below it, and the resulting global maximum 155520/577877 at $(k,n) = (6,7)$; the direct sweep of (C1)–(C3) over $\mathbb{Q}$ runs independently to $n \leq 300$. Three of the verifier's eight mutation controls target Lemma T alone: the crossover pushed down (the certificate fails at every $n_1 < 10$), the finite part truncated (6 orphan cells at $n = 9$), and (T3) run on the polynomial cap $6/n^3$ in place of $h!/n^h$ (which certifies nothing at $n = 10$). *A published earlier reading of this computation said all four failures were (C2)-only; that is false at* $(3,4)$*, where* $\gamma = 3/32 > 1/12$ *and (C1) and (C3) fail too. Three of the four are (C2)-only. The exclusion set itself is unchanged, and no statement depends on which condition fails.*

2.6 Strictness off the face

Steps (S1)–(S4) give the inequality (3) on the region (4), but not the equality case: the certified chain is tight along the whole of Ω_n, so it cannot by itself distinguish J_n/n from any other doubly stochastic matrix. The equality half comes from noticing that (S4) was proved with a margin that was thrown away when it was stated as a yes/no condition. The (C1)(C2)(C3) analysis in fact proves

$$(m/n)\sum \chi \leq \theta(n,k) \cdot D(A)/\gamma, \quad \theta(n,k) = \frac{\gamma k^2(n-1)}{(n-k)(k-1)(1-\kappa)}, \quad \kappa = \frac{3\gamma(k-2)(n-1)}{(k-1)^2}, \tag{19}$$

and (C3) is the statement $\theta \leq 1$.

> **Theorem E (strictness, with a deficit slope).** Let $(k,n) \in R$ and $A \in K_n$. Then

$$\begin{aligned} D(A) < \gamma: \quad & \Phi_k(A) \le 2 - \gamma - (1-\theta(n,k))D(A), \\ D(A) \ge \gamma: \quad & \Phi_k(A) \le 2 - D(A), \end{aligned} \tag{20}$$

and $\theta(n,k) < 1$. Consequently $\Phi_k(A) = 2 - \gamma$ forces $D(A) = 0$, i.e. $A \in \Omega_n$.

[V] `graded_verify_strict.py`: all 42 checks pass, all 8 controls fire. $\theta < 1$ on all 6899 cells of R with $n \le 120$, and on every cell of R by Lemma T, whose steps (T0)–(T5) the same verifier re-checks; the equivalence of (C3) with $\theta \le 1$ recomputed over $\mathbb{Q}$ with zero mismatches for $n \le 400$; Theorem E measured against Φ_k rebuilt from the definition at 81 off-face points, tightest slack 9.80×10^{-6}. *Two precision points are stated rather than buried.* (C3) is $\theta \le 1$, not $\theta < 1$; the strict inequality is an extra fact, and it is true — $\theta = 1$ at no cell of R. And θ is proved, not proved sharp: halving it leaves every check of the verifier passing, and the Hessian of $P_k - \gamma + \theta D$ at J_n/n restricted to $\{\sum b = 0\}$ is positive *definite*, so the bound has genuine second-order room at the equality point.

Proof. Near branch: (S1) and (S2) give $\Phi_k \le 2 - D - \gamma\,\mathrm{cap}/\mathrm{cap}_0$; then (S3), $e^{-x} \ge 1 - x$, and the sharpened (19) give $\mathrm{cap}/\mathrm{cap}_0 \ge 1 - \theta D/\gamma$. Far branch: $\Phi_k = 2 - D - P_k \le 2 - D \le 2 - \gamma$, strict when $D > \gamma$. At $D = \gamma$ exactly, equality would need $P_k = 0$, i.e. $\sigma_k(A) = 0$; by König's theorem the support then has no k-matching, so a vertex cover of at most $k-1$ lines carries all the mass n, so some line sum is at least $n/(k-1)$, whence $D \ge 4/\big(3(n-1)^2\big) > 6/n^3 \ge \gamma$ — a contradiction. Finally $D(A) = 0$ iff $r = c = \mathbf{1}$, by Maclaurin equality. ■

The slope is never small. Exactly,

$$1 - \theta = 422357/577877 = 0.7308769859\ldots \quad \text{at the worst cell } (k,n) = (6,7), \tag{21}$$

and $1 - \theta = 0.861644$ at $(7,8)$, 0.966897 at $(9,10)$, tending to 1. That $(6,7)$ is the worst cell of the whole of R, and not merely of the swept part, is Lemma T: its tail ceiling $144/955$ sits below $155520/577877$, so the maximum found in the finite part is the global one. So off the face Φ_k falls below $2 - \gamma$ by at least $0.73\,D(A)$ everywhere on R.

2.7 The face, and the theorem

On Ω_n the deficit vanishes and $\Phi_k = 2 - P_k$, so $\Phi_k = 2 - \gamma$ there is exactly $\sigma_k(A) = \binom{n}{k}^2\gamma$.

Import (Friedland 1982 [8]). For $2 \le k \le n$, σ_k on Ω_n attains its minimum $\binom{n}{k}^2\gamma(n,k)$ *only* at $A = J_n/n$.

[R], and cited alone. Friedland–Gurvits [12] Corollary 3.2 gives a second, capacity-based proof, but its equality clause is preprint-grade — the hypothesis of Theorem 3.1 there carries a typographical error, the equality sentence is ungrammatical, and the $m = n$ equality case is deferred elsewhere — so no weight is placed on it. At $k = 1$ the uniqueness genuinely fails, $\sigma_1 \equiv n$ on Ω_n; hence $k \ge 2$ throughout.

Theorem 2 (the conjecture on the capacity region). For every $(k,n) \in R$ as in (4) and every $A \in K_n$, $\Phi_k(A) \le 2 - k!/n^k$, with equality if and only if $A = J_n/n$. Quantitatively, $\Phi_k(A) \le 2 - \gamma - 0.73\,D(A)$ whenever $D(A) < \gamma$.

[V] on [R] imports (E1)–(E3) and Friedland [8]. Verifiers: `graded_verify_capfrontier.py` (29 checks, 10 controls), `graded_verify_borderrows.py` (20, 7), `graded_verify_thmb.py` (32, 9), `graded_verify_strict.py` (42, 8). *Not kernel-checked and not soon formalisable*: (E1) needs H-stable polynomials and the full Gurvits induction, and Friedland's theorem is a rock of the same size (§6.3).

What Theorem 2 does not give. A stability constant in the doubly centred directions. D vanishes identically on Ω_n, so for a perturbation with zero row and column sums the deficit of Theorem E is 0 and this route says nothing at all. The quantitative record in those directions is §5.1, which is a different machine.

3 The endpoint $k = 2$, the low lines, and the four exceptional cells

3.1 $k = 2$, every $n \ge 2$

Theorem D. $\Phi_2(A) \le 2 - 2/n^2$ for every $A \in K_n$, with equality only at J_n/n. With $\kappa = 2/(n(n-1))$ and $b = A - J_n/n$ the deficit is a manifest sum of squares on $\{\sum_{ij} b_{ij} = 0\}$:

$$(2 - 2/n^2) - \Phi_2(A) = \frac{1}{2}\big[\kappa^2\|b\|^2 + \kappa(1-\kappa)\big(\|R\|^2 + \|C\|^2\big)\big]. \tag{22}$$

Kernel-checked: `SubDittertK2.subDittert_k2`. *The statement is classical*; the contributions are the derivation from the uniform layer identity and the machine-checked proof. Since $\kappa > 0$ and $\kappa \le 1$ for $n \ge 2$, the right side vanishes only at $b = 0$, which is the equality case.

A second, independent fact pins the same line. At $k = 2$ the critical system of Φ_2 solves in closed form on the whole affine hyperplane $\{\sum a_{ij} = n\}$ — not merely on K_n — and its only solution is J_n/n: constancy of the gradient forces $a_{ij} = \alpha - \big(\binom{n}{2} - 1\big)\big(r_i + c_j\big)$, summing over j forces every r_i equal, and the mass constraint finishes it. [V]

3.2 $k = 3$, every $n \ge 3$

Theorem A ([22]). For every $n \ge 4$ and every $A \in K_n$, $E_3(r) + E_3(c) - P_3(A) \le 2 - 6/n^3$, with equality if and only if $A = J_n/n$; and more precisely

$$(2-6/n^3)-\Phi_3(A)\geq\theta_2(n)\|A-J_n/n\|_F^2,\quad \theta_2(n)=\frac{n^4+40n^2-84n+40}{n^5(n-1)^3(n-2)}>0. \tag{23}$$

With Hwang's theorem [5] at $n=3$ the line is closed at every $n\geq 3$.

Kernel-checked: `SubDittertK3.subDittert_k3_full`, all three parts in one theorem, at commit `32c811e`. *The* $n=3$ *cell is refereed literature, not Lean*: this line's support is Lean plus the published theorem, and the corollary must not be cited as a single-layer result.

3.3 $k=4$*, every* $n\geq 4$

The line is closed by three different things, and none of them inherits the grade of another. At $n\geq 10$ it is a theorem of the collar route: a violator is confined to a neighbourhood of Ω_n, a collar matrix splits orthogonally into a line-sum block and a doubly centred block, the centred block is governed by a stability form of the Tverberg–Friedland theorem, and the five cross terms between the blocks have exact reductions. At $5\leq n\leq 9$ it is five fixed-dimension certificates at anchor grade. At $n=4$ — which is the Dittert cell $(4,4)$ — it is the certificate of §5.3.

Anchor grade means an exact rational certificate accepted in full by six checks of a standalone verifier: the bound; σ_k by two structurally different algorithms; the identity by *full* coefficient comparison; definiteness of the assembled Gram matrices by exact LDL^T over $\mathbb{Q}$; equality only at J_n/n; and mutation tests rejected. Anything less is stated as what it is.

cell	bound settled on K_n	support
$(4,4)$	$\Phi_4\leq 61/32$	anchor grade; both assembled 152×152 Gram matrices positive definite over $\mathbb{Q}$; identity over 1,040 monomials (§5.3)
$(4,5)$	$\Phi_4\leq 1226/625$	anchor grade; 26 exact LDL^T factorisations of size 350
$(4,6)$	$\Phi_4\leq 107/54$	anchor grade; identity over 40,386 coefficients, cone size 702
$(4,7)$	$\Phi_4\leq 4778/2401$	anchor grade; 156,555 coefficients, positivity by congruence
$(4,8)$	$\Phi_4\leq 1021/512$	anchor grade; 496,448 coefficients, cone size 2144
$(4,9)$	$\Phi_4\leq 4366/2187$	anchor grade; 1,355,805 coefficients, cone size 3402
$n\geq 10$	$\Phi_4\leq 2-24/n^4$	Theorem H of [22]: written proof plus the exact verifier `graded_verify_k4.py`; *not* kernel-checked

Table 3: The line $k=4$. In every row equality holds only at J_n/n.

3.4 The four exceptional cells

The cells excluded from (4) are $(k,n)=(3,4),(3,5),(4,5),(5,6)$. The first two are $k=3$ cells and fall under §3.2; the third is $(4,5)$ in Table 3. The fourth is an anchor of its own.

The $(5,6)$ anchor. $\Phi_5(A)\leq 643/324=2-5!/6^5$ on K_6, with equality only at $J_6/6$.

`[V]`, anchor grade, six checks. The companion cells $(5,5)$ — which is also the Dittert cell $n=5$ — and $(5,7)$ are recorded in §5.3 and in the data kit; no statement here rests on $(5,7)$.

So the whole of $2\leq k\leq n-1$ is covered, at the grades of Table 2. One line remains.

4 The line $k=n$: Dittert's conjecture

This is the line whose grade the whole assembly inherits, and it is stated here with its provenance rather than with its verdict.

n	source of record	grade	what carries it
2	Sinkhorn 1984 [4]	[R]	refereed
3	Hwang 1987 [5]	[R]	refereed
4	our anchor certificate	[V]	six checks; both assembled 152×152 Gram matrices positive definite over $\mathbb{Q}$. *No refereed proof exists; the first public claim is the July 2026 assembly's [20]*
5	our anchor certificate	[V]	six checks; both assembled 350×350 Gram matrices positive definite over $\mathbb{Q}$
$6 \le n \le 15$	Lu, revision 2.0 [17]	[P]	every analytic step re-derived by hand, every constant reproduced as an exact rational, 98 four-variable certificates re-certified, 3 Bernstein trees node for node, 129 exact comparisons across $8 \le n \le 15$ with zero failures
16	Kafidov [19]	[P]	full re-derivation; all exact rational bounds reproduce
≥ 17	Pang [18]	[P]	full re-derivation; every constant reproduces exactly

Table 4: The $k = n$ line. [V] means *we* checked it, not that anyone refereed it; [P] means an unrefereed public source whose mathematics we re-derived. Dittert's conjecture is described in the current refereed literature as unsettled, and this table does not change that.

Priority at $n = 4$, stated plainly. The cell $n = 4$ has no refereed proof — Cheon–Wanless [3] p. 792 records it as "still open for $n \ge 4$", and Pang's Remark 2 repeats it. It is nonetheless *not first here*: the July 2026 public assembly [20] already contains an $n = 4$ certificate, and that is the first public claim. The certificate of §5.3 is **independent confirmation at anchor grade, not priority.** The same wording applies at $n = 5$.

4.1 Why the capacity chain cannot reach $k = n$

At $k = n$ one has $m = 0$, $\mathrm{cap}_0 = 1$, $M_n(A) = A$, and Theorem H′ is a literal no-op: $C_{\mathrm{new}}(n, n) = C_{\mathrm{ref}}(n, n) = n!/n^n$ exactly, [V] for $n = 2...44$. Both facts that make the line look reachable — $\rho_{\mathrm{ref}}(n, n) = 1$ and $G(n, 0) = 1$ — are true and neither helps. The reason is a theorem, not a missing estimate.

> **Lemma K1.** For every $A \in K_n$, $\mathrm{cap}(A) \le \prod_i r_i \le 1$, and $\mathrm{cap}(A) = 1$ *if and only if* $A \in \Omega_n$.

So step (S3) at $k = n$ asserts $\mathrm{cap}(A) \ge 1$, which is *exactly the hypothesis* $A \in \Omega_n$: not a weak estimate but a false one at every off-face point. This is what the border was doing all along — at $m \ge 1$ the witness of §2.5 dumps the line-sum deficit into the border block; at $m = 0$ there is nowhere to put it. (Step (S4) at $m = 0$ is the vacuous $0 \le D/\gamma$, so the conditions (C2), (C3) are symptoms and not the break.) On Ω_n itself the chain is exact and refereed: $\mathrm{per}(A) \ge \gamma$ with equality only at J_n/n. The [P] grade attaches to $K_n \setminus \Omega_n$ — which is the whole of Dittert's difficulty, so this is a restatement and not a reduction.

Replacing the witness by the exact capacity reduces all of $k = n$ to one statement, and that statement is false.

> **Theorem K.** The inequality $\gamma(1 - \mathrm{cap}(A)) \le D(A)$, for $A \in K_n$ with $D(A) < \gamma$, is FALSE at every $n \ge 2$, with exact rational witnesses that are strictly positive, fully indecomposable, and at which the capacity infimum is attained.
>
> **Theorem K′ (the sharp criterion).** Near $A_0 \in \Omega_n$ that inequality holds *if and only if* $\gamma \le 1 - \sigma_2(A_0)^2$, with σ_2 the second largest singular value.
>
> [V] `graded_verify_kn.py`: all 35 checks pass, all 9 controls fire. Witnesses $A(n, \varepsilon, t) = (1 - \varepsilon)I_n + (\varepsilon/n)J_n + t(E_{12} - E_{11})$ with $D = t^2$ exactly, $\varepsilon = \gamma/5$, at $n = 2...7$; the measured threshold is $\varepsilon^* = 1 - \sqrt{1 - \gamma}$ to nine places. The witnesses are strictly positive, so every column degree is n and the degree-refined form of (E1) gives exactly $n!/n^n$: *the refutation survives the strongest form of the imported engine.*

Since $\sigma_2 \to 1$ at every permutation matrix and at every decomposable point, the criterion fails on an open neighbourhood of the permutation matrices at every $n \ge 2$. The structural reason is worth stating plainly: $\mathrm{cap} \equiv 1$ on *all* of Ω_n, while Dittert's own margin $\mathrm{per}(A) - \gamma$ ranges over the whole of $[0, 1 - \gamma]$ there. A route whose value on the equality manifold is constant cannot tell J_n/n from I_n, so it must survive the worst off-face capacity decay uniformly over that manifold — and that decay is unbounded. Consistently with this, no analogue of Theorem E exists at $k = n$ with any $\theta < 1$: the deficit there is $\gamma(\mathrm{cap} - 1) \le 0$.

4.2 The reduction: what would make this line self-contained

Step (S2) is lossy in exactly one identifiable way. The exact statement is $\mathrm{per}(A) = \mathrm{cap}(A)\,\mathrm{per}(B)$ for the Sinkhorn scaling $A = D_1 B D_2$ with $B \in \Omega_n$, and (S2) replaces $\mathrm{per}(B)$ by its van der Waerden floor γ. Keeping $\mathrm{per}(B)$, the requirement holds locally at *every* face point: $\mathrm{per}(B) - \gamma > 0$ off J_n/n, and at J_n/n the two vanishing orders match.

> **Theorem R (the reduction).** The missing input at $k = n$ is a *quantitative van der Waerden stability estimate* — $\mathrm{per}(B) - n!/n^n$ bounded below in terms of $1 - \sigma_2(B)^2$ for $B \in \Omega_n$ — and not a capacity estimate at all. Any such estimate, composed with Theorem K′, makes the line $k = n$ self-contained at the grade of §2.
> [V] for the factorisation and the composition arithmetic (`graded_verify_kn.py`). **No such estimate exists in the literature.** A search of the Gurvits capacity line, Brändén–Leake–Pak, Schrijver, Linial–Samorodnitsky–Wigderson, the quasirandom Latin square line and the Dittert-specific literature returned nothing of the form $\mathrm{per}(B) \geq n!/n^n + c \cdot \mathrm{dist}(B, J_n/n)^2$ or any σ_2-indexed variant; and *no capacity-based approach to Dittert appears anywhere.* The successful classical route uses a pair of *structural* theorems instead — Knopp–Sinkhorn [16] on the zero face and Hwang 1986 [6] in the positive-support case, with the Cheon–Wanless dilation between them — and neither is indexed by σ_2.

Two cheap versions of the missing half are recorded as closed. The exact factorisation $\mathrm{cap}(A)\,\mathrm{per}(B) \geq \gamma - D(A)$ is *literally* Dittert's conjecture, checked as an identity of statements, so all content lies in bounding the two factors separately — and Theorem K says the trivial choice fails. And the scale-invariant strengthening of Gurvits that would give Dittert in one line,

$$(\mathrm{S}') \quad \mathrm{per}(A) \geq (n!/n^n) \prod_i r_i \prod_j c_j \quad \text{for every } A \in K_n, \tag{24}$$

is false, and false inside $\{D < \gamma\}$, the only region that matters: at $n = 3$, with A the matrix with rows $(0, 1/2, 1/6)$, $(1/2, 0, 5/6)$, $(1/6, 2/3, 1/6)$, one has $p = 8/9$, $q = 49/54$, $D = 11/54 = (11/12)\gamma < \gamma$ and $\mathrm{per}(A) = 1/6 < \gamma pq = 392/2187$, a ratio of 0.929847. [V] exactly over $\mathbb{Q}$; Dittert itself holds there with margin 4/27. (S′) is tight on the whole rank-one family and at J_n/n, which is why it is worth refuting rather than ignoring.

5 Results of independent interest

Three by-products stand on their own, and one of them supplies exactly what §2 cannot.

5.1 Unconditional thresholds, with stability

The capacity route gives no constant in the doubly centred directions. A different machine does: confinement traps a violator in a collar of the Birkhoff polytope, a uniform stability form of the Tverberg–Friedland theorem governs the centred block, and a layer budget prices every cross term.

> **Theorem U (thresholds with stability).** For every $k \geq 3$ and every $n \geq \tilde{N}(k)$, every $A \in K_n$ satisfies $\Phi_k(A) \leq 2 - k!/n^k$, with equality only at J_n/n, and quantitatively
> $$(2 - \gamma) - \Phi_k(A) \geq c_U(n,k) \|A - J_n/n\|_F^2, \quad c_U(n,k) > 0 \text{ explicit}. \tag{25}$$
> The η-priced thresholds
> $$\tilde{N}(5 \ldots 12) = 29, 35, 43, 53, 63, 75, 88, 102 \tag{26}$$
> are *unconditional.*
> [V], composed from parts of mixed grade, and the composition is the object verified (`graded_verify_universal.py`, 301 checks; `graded_verify_uniformG.py`; `graded_verify_collar.py`). Three links are kernel-checked — the layer identity, confinement, and Newton/Maclaurin — and the rest are written proofs with exact verifiers behind them. The thresholds are unconditional because the connected-cumulant bound $|S_G(B)| \leq Q$ is now proved for every pattern of at most twelve edges (§5.2); the range $k \geq 13$ remains conditional on one further core lemma. $\tilde{N}(k)$ is a threshold *for this argument* and is not claimed optimal.

This is the campaign's only quantitative-stability record, and it is the one place where the near-diagonal corner is *not* reached: the wedge $k + 1 \leq n < \tilde{N}(k)$ is untouched by Theorem U and is covered, in this paper, by §2 instead. The two machines are complementary in exactly that sense.

5.2 The core census and L-CUT

The obstruction to the local calculus behind Theorem U is a combinatorial object that appears to be new: a *core*, a simple graph of minimum degree ≥ 3, first realised at edge count $e_{\min}(H) = 2e(H) - \mathrm{maxcut}(H)$.

> **Theorem (census).** There are exactly eleven cores realisable at $e \leq 12$, and exactly forty-one at $e \leq 14$. The first 3-connected cores are the wheel W_4 at $e = 10$ and the prism at $e = 11$ — *not* the cube.
>
> **Theorem (L-CUT at $e \leq 14$).** One inequality schema subsumes the four separate core lemmas: split a core at a separator S with $V - S = A \sqcup B$, pay the edges inside S pointwise, and bound each side in ℓ^2. The covering condition $N(A) - A = N(B) - B = S$ is not a convenience but is *forced* by the ℓ^2 step. Of the forty-one cores at $e \leq 14$, exactly two are complete and the other thirty-nine admit a covering split with both sides at most two vertices, in only three side shapes; K_4 is a wheel, so K_5 is the unique hole, and it is first realised at $e = 14$.

> [V] `graded_verify_u5core3.py` (727 checks, 6 controls firing at 30 positions) and `graded_verify_canon.py` (2106 checks, 4 controls at 12 positions); the sweeps at $e = 8...12$ certify $127, 317, 1018, 2989, 9930$ isomorphism classes with none uncertified, which is what makes $e_0 = 12$ and the thresholds above unconditional. A separate delineation result: the bound is *false* without the entry bound $b_{ij} \geq -1/n$, with an explicit nine-edge witness whose core is empty — so the failure is in leaf propagation, not core structure. This material may be published separately; it is stated here at pointer level.

5.3 Dittert at $n = 4$ and $n = 5$

Both cells carry exact sum-of-squares certificates produced and verified here, at the full six-check anchor standard, including equality only at J_n/n.

cell	what was produced	what checked it
$(5,5)$	440 unknowns, 87 rows, 21 canonical blocks; strictly feasible at round 0, least block eigenvalue $+1.913166 \times 10^{-5}$	identity over 14,005 monomials, full coefficient comparison over $\mathbb{Q}$; both assembled 350×350 Gram matrices positive definite over $\mathbb{Q}$, least pivots 1.1546×10^{-4} and 3.1719×10^{-5}
$(4,4)$	423 unknowns; the 21-block design *does not exist* at $n = 4$, so the certificate came from the early-exit arm	identity over 1,040 monomials; both assembled 152×152 Gram matrices positive definite over $\mathbb{Q}$, with no multiplicity count anywhere in the chain

Table 5: The two Dittert cells settled here. Independent confirmation of the July 2026 assembly's $n = 4$ claim, not priority.

Two details are load-bearing for the grade. The check that distinguishes $k = n$ from $k = n - 1$ cannot be the bound itself: $k!/n^k = (k-1)!/n^{k-1}$ exactly when $k = n$, so at $(5,5)$ the tested bound $1226/625$ is also $2 - 4!/5^4$ and carries no k-information whatever. The discriminating evidence is the coefficient count — 14,005 monomials at $(5,5)$ against 7,875 at $(5,4)$ — typed in from an independently derived closed form *before* the run, the same closed form having reproduced 7,875, 40,386 and 809,529 at three other cells. A $k = 4$ certificate could not have passed. And the cell data at $(5,5)$ came from a cache that predated the work, so it was rebuilt from scratch and compared: identical to the fresh $k = 5$ build, and different from the $k = 4$ cell in 3 of 87 constraint rows and 18 of 87 right-hand sides.

6 Verification

6.1 What a graded verifier is

Every displayed constant in this paper is produced by a verifier that computes in exact rational arithmetic end to end, with no floating-point number entering any decision — floats appear in format strings and nowhere else. That is necessary and not sufficient, so each verifier additionally carries *fault-injection controls*: deliberately mutated inputs or disabled lemmas that the verifier is required to reject at two or more independent positions, while staying silent on clean input. A verifier with no firing control is treated as unverified.

Two culture rules are stated because they were each learned from a failure. *A mutation control that disables a lemma while a synonym stays enabled tests nothing*: controls must partition the hypothesis space, not the code. And *a checker needs its own control*: several counting instruments here were caught mis-counting by a positive control that measured a known answer.

6.2 One command

The whole campaign re-verifies with a single command, which discovers and runs every graded verifier in sequence and exits non-zero if any fails.

```
cd problems/permanents/sub-dittert
../guard.sh python3 verify_all.py
```

Twenty-two verifiers are discovered, and the full suite is green. The run of 3 August 2026 reports `VERIFIERS: 22/22 pass` with `OVERALL: ALL VERIFIERS PASS`, every verifier returning zero failures; the log is `verify_all.log` in the reproduction kit. The verifiers carrying the theorems of this paper are the following; each count is quoted from that verifier's own log.

verifier	what it certifies	result
`graded_verify_capfrontier.py`	Theorem F and the exact price of each further line	29 checks, 10 controls fire
`graded_verify_borderrows.py`	Theorems G′/H′; the extraction bound (9) re-derived from Newton; the degeneration at $m = 0, 1$	20 checks, 7 controls fire
`graded_verify_thmb.py`	the entropy witness, its exactness on the face, and (C1)(C2)(C3)	32 checks, 9 controls fire
`graded_verify_strict.py`	Theorem E, the König corner, $D = 0$ iff on Ω_n, and Lemma T — steps (T0)–(T5), the crossover $n_1 = 10$ and the 21 exact cells below it	42 checks, 8 controls fire
`graded_verify_kn.py`	Theorems K and K′, the σ_2 threshold, the refutation of (S′)	35 checks, 9 controls fire
`graded_verify_u5core3.py`	the census, the four core lemmas, L-CUT	727 checks, 6 controls at 30 positions
`graded_verify_canon.py`	the sweeps establishing $e_0 = 12$	2106 checks, 4 controls at 12 positions
`graded_verify_universal.py`	the composed Theorem U and its interface	301 checks, 0 failures

Table 6: The verifiers behind this paper's displayed constants. First acceptance of any certificate goes through a separate trusted verifier that shares no code with the producer; the modular fast layer is for re-verification only.

6.3 What is kernel-checked, and what is not

> The Lean 4 development elaborates with no `sorry`, no declaration depends on `sorryAx`, and `native_decide` is used nowhere. Every declaration cited in support of a stated result carries a `#print axioms` command returning a subset of `[propext, Classical.choice, Quot.sound]`. Build success is not an axiom audit, and the two are kept separate.

Kernel-checked. The line $k = 3$ in full — inequality, equality case and the stability bound $\theta_2(n)$ — as `subDittert_k3_full` at commit `32c811e`, whose audit block covers all 377 named declarations of its file. The line $k = 2$ (`subDittert_k2`). The uniform layer identity at every $k \le n$ (`universal_identity`). Confinement at every k, with its Maclaurin hypothesis discharged (`theoremM'`, `confinement'`). Newton's inequalities and Maclaurin for every real-rooted real polynomial (`newtonAt_all`, `newton_esymF`, `pnorm_le_two`), neither of which is in Mathlib v4.14.0. The Tverberg–Friedland stability form at $k = 2, 3, 4, 5$ over the whole range each case states. The padding refutation, and the shifted-coefficient framework.

Not kernel-checked, and not soon formalisable. Everything in §2. The obstruction is external, not internal: the five internal steps of the chain are all within reach of Mathlib — weighted AM–GM, an exact logarithm identity, Newton, Maclaurin, and three decidable rational conditions with two monotone ratio lemmas — but the input the cell rests on is Gurvits' capacity theorem (E1), which needs H-stable polynomials and the full Gurvits induction, and the equality case rests on Friedland's theorem. Those are the two large unformalised rocks, together with Egorychev–Falikman behind them. Any Lean statement of Theorem 2 today would have to carry (E1) as a named hypothesis, which is the honest shape and is queued at low priority.

Also not kernel-checked. The $k = 4$ collar theorem and every fixed-dimension certificate: those rest on written proofs plus exact standalone verifiers with stored rational witnesses. The $k = n$ line is not formalised at any n. What the kernel cannot check anywhere is that the Lean definitions say what the 1992 paper says; two validation theorems argue that they do — $\Phi_k(J_n/n) = 2 - k!/n^k$ for every $k \le n$, and $\sigma_k(xy^T) = k!e_k(x)e_k(y)$, which is the check that rows and columns are read from *different* subsets — and that is the only place left where a human error would survive the machine.

7 Two conjectures

7.1 The second extremal structure

Three independent searches, run with different objectives, all located the same second-highest critical structure of Φ_k on K_n: an exhaustive Karush–Kuhn–Tucker census over structured faces, the layer analysis of the flow to J_n/n, and the equality family of the diagonal boundary gap. In each the answer is the same orbit — the 2-regular circulants $A_n^{(t)} = (I_n + C^t)/2$ with $\gcd(t, n) = 1$, all equivalent under row and column permutation to $A_n = (I_n + C)/2$.

The evidence is exact. The bipartite support graph of $I + C$ is a single $2n$-cycle, so deleting one row and one column leaves two even paths with a unique perfect matching each; hence $\mathrm{per}(A_n(p|q)) = 2^{1-n}$ for *every* (p, q), $\mathrm{per}(A_n) = 2 \cdot 2^{-n}$, and $\sigma_{n-1}(A_n) = n^2 2^{1-n}$. Consequently

$$\Phi_n(A_n) = 2 - 2^{1-n}, \quad (2 - n!/n^n) - \Phi_n(A_n) = 2^{1-n} - n!/n^n > 0 \quad \text{for every } n \geq 3, \tag{27}$$

with values $1/36, 1/32, 241/10000, 41/2592, 71569/7529536, 709/131072$ at $n = 3...8$ and equality at $n = 2$, where $A_2 = J_2/2$. Of the 784 structured faces tested at $4 \leq n \leq 7$, exactly 14 satisfy the first-order conditions, and every one of them has $k = n$ and is in this orbit; at $k < n$ the same matrices fail with an exact positive rational witness ($1/12$ at $(k, n) = (2, 4)$, $1/8$ at $(3, 4)$, $11/120$ at $(4, 6)$). [V]

> **Conjecture 1 (the second extremum).** For every $n \geq 3$, the largest value of Φ_n on $K_n \setminus \{J_n/n\}$ is attained at $A_n = (I_n + C)/2$, and equals $2 - 2^{1-n}$. Equivalently, the tightest non-J margin is exactly $2^{1-n} - n!/n^n$, which is $1/36$ at $n = 3$.
>
> Conjecture grade; the evidence above is exact and the search coverage is not. Its practical content: *every remaining sharpness question in this problem localises at that orbit*, and any future route to $k = n$ must clear $1/36$ at $n = 3$. At $k < n$ the orbit is not critical, which is consistent with §2's chain being strict off the face.

7.2 Quantitative van der Waerden stability

§4.2 showed that one estimate would make the $k = n$ line self-contained, and that no such estimate is in the literature at any strength.

The estimate is stated *qualitatively*, and deliberately so. Two exact facts fix its shape before any constant is written down, and both are cheap enough that stating a constant without them would be careless.

> **Conjecture 2 (stability of the van der Waerden minimum).** For every $n \geq 3$ there is a constant $c(n) > 0$ such that for every $B \in \Omega_n$,
>
> $$\operatorname{per}(B) - \frac{n!}{n^n} \geq c(n) \cdot \frac{n!}{n^n} \cdot \sigma_2(B)^2, \tag{28}$$
>
> with σ_2 the second largest singular value of B.
>
> Conjecture grade; no value of $c(n)$ is claimed, and no claim is made that $c(n)$ can be taken absolute. Both sides vanish at J_n/n and to the same order: writing $B = J_n/n + X$ with X doubly centred, $\sigma_2(B) = \|X\|_{\text{op}}$ exactly, and $\operatorname{per}(B) - n!/n^n = (n!/n^n)\frac{n}{2(n-1)}\|X\|_F^2 + O(\|X\|^3)$, so the rank-one directions cap $c(n) \leq n/(2(n-1))$. *Nothing of this shape appears in the refereed literature*; the known structural substitutes — Knopp–Sinkhorn [16] on the zero face, Hwang [6] on the positive-support case — are not indexed by σ_2.

Two exact checks, and one refutation that fixes the index. At the circulant $A_n = (I_n + C)/2$ of §7.1 the margin is exactly the tight value of Conjecture 1, so (28) is a genuine constraint there and not a formality: at $n = 3$, per $= 1/4$, $\sigma_2^2 = 1/4$ and the margin is $1/36$, forcing $c(3) \leq 1/2$; at $n = 4$, per $= 1/8$, $\sigma_2^2 = 1/2$ and the margin is $1/32$, forcing $c(4) \leq 2/3$. Both are consistent with the local cap $n/(2(n-1)) = 3/4, 2/3$, so the conjecture is not refuted at either cell, and the $1/36$ point of §7.1 is respected. [V] exactly over $\mathbb{Q}$. *The index cannot be* $1 - \sigma_2^2$. That variant — $\operatorname{per}(B) - n!/n^n \geq c(n!/n^n)(1 - \sigma_2(B)^2)$ — reads correctly at the permutation matrices but is false at $B = J_n/n$ for *every* $c > 0$ and every n, since there the left side is 0 while $\sigma_2 = 0$ makes the right side $c\,n!/n^n > 0$. It is worth recording because it is the form Theorem K′ appears to ask for.

Remark (the quantitative demand, and why (28) does not meet it). What the composition of §4.2 actually consumes is a *pair* of bounds, $\operatorname{cap}(A) \geq \Phi(D(A), \sigma_2(B))$ and $\operatorname{per}(B) \geq \gamma\Psi(\sigma_2(B))$ with $\Phi \cdot \gamma \cdot \Psi \geq \gamma - D$. The second-order law behind Theorem K′ prices Φ: along the worst direction at $A_0 \in \Omega_n$ the capacity drop is $D/\big(1 - \sigma_2(A_0)^2\big)$ to second order, so Φ degrades like $1/(1 - \sigma_2^2)$ in exactly the regime that decides the line, and Ψ would have to compensate by *growing* as $\sigma_2 \to 1$. Estimate (28) does not do that: its right side stays bounded by $(n!/n^n)n/(2(n-1))$. So (28) is the correct *shape* for a stability estimate — right index, matched vanishing order at J_n/n, exact caps at $n = 3, 4$ — but it is not by itself enough to close $k = n$, and the paper claims no more than that. The two factors are anti-correlated precisely where each is extreme, which is the structural reason §4.1 gives for the line being closed to this route; a sufficient Ψ would have to be a genuinely stronger statement than (28), and we do not have one.

8 What is not claimed

The plane claim (3) carries [P] from the $k = n$ line and should be quoted with that tag, not one hop away from it. Theorem 2 is inequality *and* equality on the region (4), but it supplies no stability constant in the doubly centred directions at any cell; §5.1 supplies those only for $n \geq \tilde{N}(k)$, and no constant is claimed optimal anywhere. The constant θ of Theorem E is proved, not proved sharp. Nothing in §2 is kernel-checked, and the reason is external (§6.3). No floating-point number enters any accepted statement. Priority is claimed nowhere on the $k = n$ line; the $n = 4$ and $n = 5$ certificates are independent confirmation. Finally, nothing here is refereed, including this paper, and priority claims of any kind are perishable and should be re-checked before submission anywhere.

A catalogue of closed strategies is maintained alongside this work and is part of the architecture rather than an appendix curiosity: eleven named strategy classes are refuted with exact witnesses — balancing, k-monotonicity, padding transfer (machine-checked), the iterated lift, real-rootedness, the odd-layer lemma, the d-vector flow, ratio monotonicity (which is Holens–Đoković, false by Wanless's 1999 witness), first-order maximiser localisation, the cumulant bound without its entry hypothesis, and generic-target dimension counting. Several of these sharpen or correct statements in the published literature, and each closes a seam that would otherwise be re-entered.

9 Data availability

The exact rational certificates and their standalone verifiers, the graded verifiers with their fault-injection controls and logs, the audit rebuilds behind Table 4, the census sweeps behind §5.2, the refutation witnesses named above, and the Lean development at the commits cited in §6.3 are supplied as a reproduction kit, whose README names the claim each file backs and the procedure for re-running each check. The failed design branches are retained there, since several closed routes are only checkable against them.

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